Study of Different Mechanical Properties of Woven Jute Fiber Reinforced Composites and It's Application as a Roofing Sheet

A project report is submitted to the Department of Mechanical Engineering, Khulna University of Engineering & Technology (KUET), and Bangladesh in partial fulfillment of the requirements for the Special Studies (ME 3200) course.

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- Author

ABSTRACT

Today's world is seeing it's future in composite material. Fiber reinforced composite has achieved great importance in aerospace & automobile engineering specially because of it's high strength and light weight. Bangladesh is enriched with a lot of natural fibers. To find the applications of these fibers as hybrid composites in product manufacturing, research work should be done. The use of jute fibers as reinforcement for development of composite materials has increased in nowadays because the vexation shown by the environment due to increase in fuel prices, depletion of fossil fuels, global warming are the major concerns which force the researchers to work in the area of green composites. The jute fiber reinforced composites also replace old materials such as steel and wood. The present work is on the development of roofing sheet material using woven jute fiber Reinforced as composite material and flexural tests will be experimented. Fabrication methods of different form of jute fiber reinforced composites have been studied.

1.0 Introduction:

The advancement of mankind is defined in terms of the development in materials i.e. the Stone Age, the Bronze Age, and the Iron Age. The recent era of material belongs to the composite materials because of its numerous number of advantages. The word 'composites' derived from the Latin word composites, which means put together signifying something made by putting together different parts or materials [1]. A composite is a material made when two or more different materials are combined together to create a superior and unique material. Composite material consists of strong carry-load materials which are embedded in a weaker material. The stronger material is commonly referred as reinforcement and the weaker material is referred as matrix. The two metals works together to give the composites unique properties. However, within the composites the material do not dissolve or blend into each other. The reinforcement provides the strength and rigidity which helps to support the structural load. The matrix or the binder helps to maintain the position and orientation of the reinforcement and is somewhat more brittle. The reinforcement materials such as Fibers are thin and but integrity is not maintained. In matrix materials, strength values are less and hence fibers or matrix alone cannot find its application as a structural material but when these two materials are combined we get a composite materials which is light weight, stiff, strong and tough [2]. Composite materials from man-made fiber (i.e. glass fiber, carbon fiber etc.) are already available as products for consumer and industrial uses. A relatively newer concept is to consider natural fiber as a reinforcing material [3]. The growing ecological concern and governmental rules lead to rise in the demand of the natural fibers as a substitute of synthetic fibers. The natural fibers such as hemp, sisal, jute, flax and bamboo are renewable and biodegradable in nature and possess high technical qualities such as good modulus and specific strength, low density and cost, and reduced dermal and respiratory irritation. The mechanical properties of natural fibers, particularly hemp, sisal, flax, and jute are relatively good, and may compete with glass fiber in terms of specific strength and modulus. Interestingly, numerous types of natural fibers which are abundantly available have proved to be effective and good reinforcement in the thermoplastic and thermoset matrices. So researching on natural fiber composites is very important. Jacob Olaitan Akindapo, Umar Alhaji BINNI, Olawale Monsur have made roofing sheet by using groundnut sell particles and epoxy resin as composite material. They have tested different tests like water absorptivity test, flexural strength, tensile Strength, impact strength [4]. Jochen Gassan and Andrzej K. Bledzki have investigated different properties of a jute-polypropylene composites. They also worked on improving different properties like tensile and impact strength [5]. Madhukiran.J, Dr.T.Venkateswara Rao, Dr.S.Madhusudan, Dr.

R.Umamaheswara Rao have worked on jute-coir hybrid natural fiber composites. They have done different tests like tensile test, flexural tests and shown that jute-coir fiber hybrid composites exhibited superior properties when compared to pure composites [6]. So, several research have been done on different natural fiber composites. But still there are different kinds of fibers which properties are not uncovered to us.

2.0 Objectives:

The specific objectives of the study work are:

(i) To study about natural fiber composites.
(ii) To study about fabrication processes of jute composites as roofing sheet.
(iii)To study of mechanical behavior of jute composite.

3.0 Historical Background:

The earliest man-made composite materials were straw and mud combined to form bricks for building construction. Ancient brick-making was documented by Egiptian tomb paintings [7]. Natural cement-stones Papier-marchie a composite of paper and glue, has been used for hundreds of years. The first artificial fiber reinforced plastic was bakelite which dates to 1907, although natural polymers such as shellac predate it. Natural cement-stones Papier-marchie a composite of paper and glue, has been used for hundreds of years. The first artificial fiber reinforced plastic was bakelite which dates to 1907, although natural polymers such as shellac predate it. Polywood 3400 BC by the Ancient Mesopotamians; gluing wood at different angles gives better properties than natural wood. Cartonnage layers of linen or papyrus soaked in plaster dates to the first intermediate period of Egypt c. 2181–2055 BC and was used for death marks. Cob (material) Mud Bricks, or Mud Walls, (using mud (clay) with straw or gravel as a binder) have been used for thousands of years [8]. Concrete was described by Vitruvius, writing around 25 BC in his Ten Books on architecture distinguished types of aggregate appropriate for the preparation of lime mortars. For structural mortars, he recommended pozzolana which were volcanic sands from the sand like beds of Pozzouli brownish-yellow-gray in color near Napies and reddish- brown at Rome. Vitruvius specifies a ratio of 1 part lime to 3 parts pozzolana for cements used in buildings and a 1:2 ratio of lime to pulvis Puteolanus for underwater work, essentially the same ratio mixed today for concrete used at sea. Natural cement-stones Papier-marchie a composite of paper and glue, has been used for hundreds of

years. The first artificial fiber reinforced plastic was bakelite which dates to 1907, although natural polymers such as shellac predate it. One of the most common and familiar composite is fiber glass in which small glass fiber are embedded within a polymeric material (normally an epoxy or polyester). The glass fiber is relatively strong and stiff (but also brittle), whereas the polymer is ductile (but also weak and flexible). Thus the resulting fiberglass is relatively stiff, strong, flexible, and ductile.

3.1 Concept of composites:

A composite material is made by combining two or more materials – often ones that have very different properties. The two materials work together to give the composite unique properties [9]. Composite material can be well-defined as an amalgamation of two or more than two materials (reinforce, fillers, and binder) different in composition on a very small scale. Composite materials (also called composition materials or shortened to composites) are materials made from two or more than two constituents or materials with considerably differ in physical and chemical properties, that when amalgamated, make a material with appearances different from the individual components. We can say that do not lose their individual identities but still impart [10]. Composite material consists of strong carry-load materials which are embedded in a weaker material. The stronger material is commonly referred as reinforcement and the weaker material is referred as matrix. Within the composites the material do not dissolve or blend into each other. The reinforcement provides the strength and rigidity which helps to support the structural load. The matrix or the binder helps to maintain the position and orientation of the reinforcement and is somewhat more brittle. The reinforcement materials such as Fibers are thin and but integrity is not maintained. Reinforcements for the composites can be fibers, fabrics particles or whiskers. Fibers are essentially characterized by one very long axis with other two axes either often circular or near circular. Particles have no preferred orientation and so does their shape. Whiskers have a preferred shape but are small both in diameter and length as compared to fibers. Figure 2.1 shows types of reinforcements in composites.

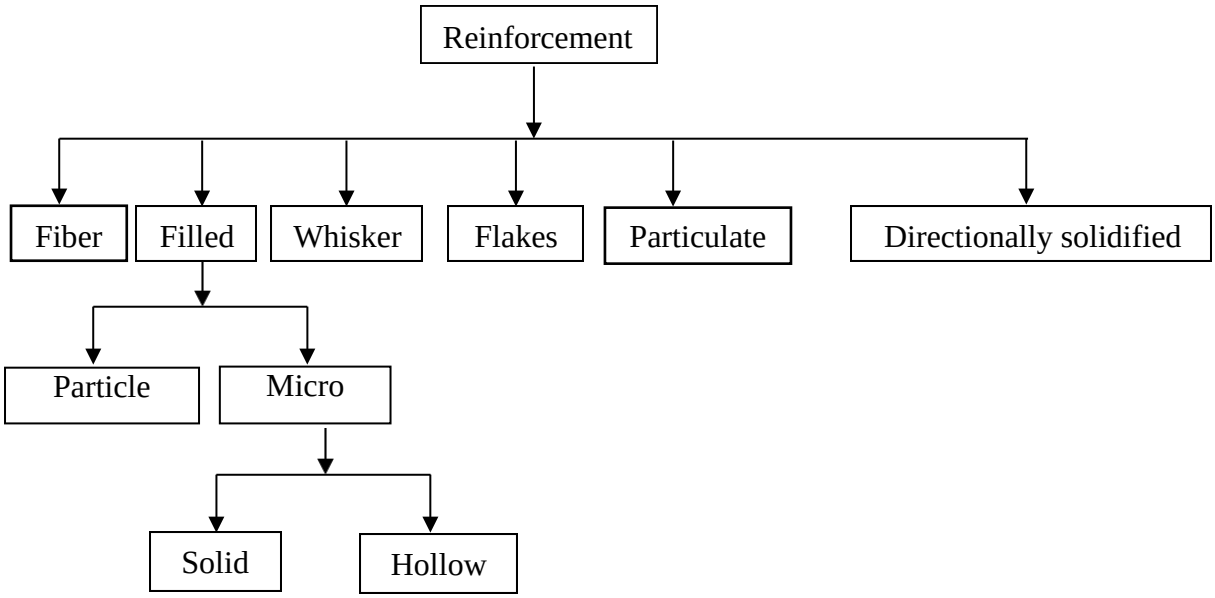


Fig. 2.1: Types of reinforcements

In matrix materials, strength values are less and hence fibers or matrix alone cannot find its application as a structural material but when these two materials are combined we get a composite materials which is light weight, stiff, strong and tough. [Basic can be taken] We can get some knowledge about the classification of matrix material from figure no 2.1.

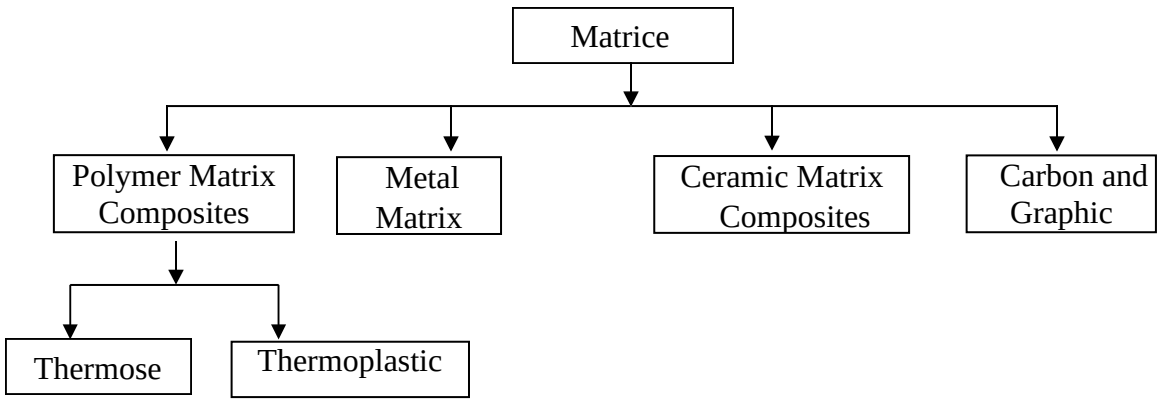


Fig. 2.2: Classification of matrix material

3.2 Polymer Matrix Composites (*PMC*):

A polymer matrix composite (*PMC*) is a composite material composed of a variety of short or continuous fibers bound together by an organic polymer matrix. *PMC*s are designed to transfer loads between fibers through the matrix. Two main kinds of polymers are thermosets and thermoplastics. Thermosets have qualities such as a well-bonded three-dimensional molecular structure after curing. They decompose instead of melting on hardening.

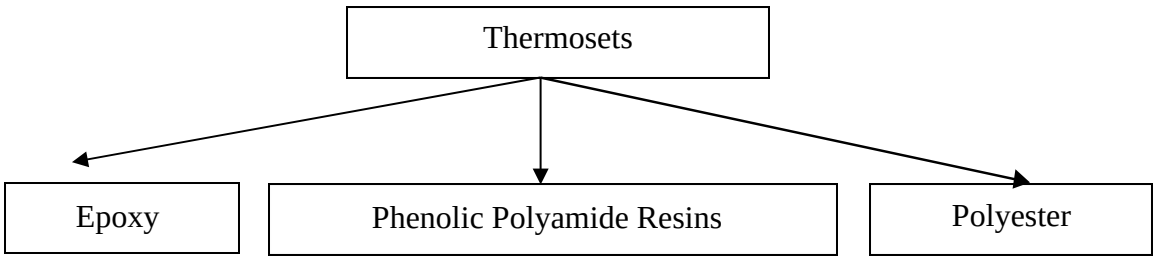


Fig. 2.3: Thermosets Material Types

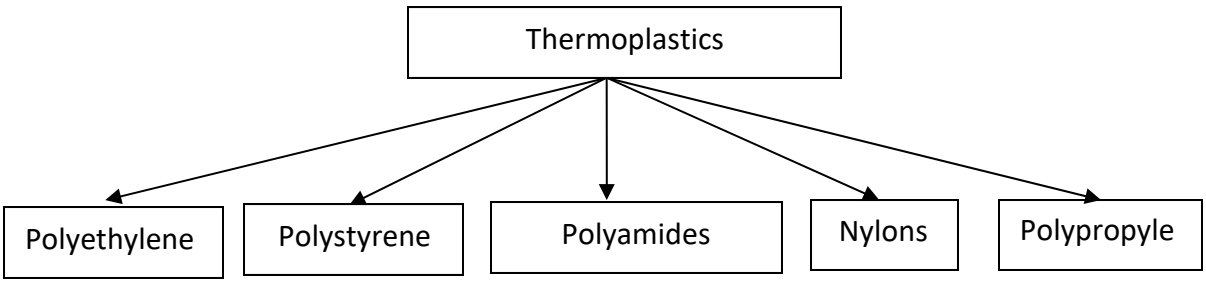


Fig. 2.4: Thermoplastic Material Types

3.3 Types of composites:

Various types of composites are used in the industries:

a) **Particulate Composites:** A composite that consists of tiny particles of one material embedded in another material. The particulates can be very small particles (< 0.25 microns), chopped fibers (such as glass), platelets, hollow spheres, or new materials such as Bucky balls or carbon Nano-tubes.

b) **Flake Composites:** A flake composite consists of thin, flat flakes held together by a binder or placed in a matrix. Two common flake materials are glass and mica. (Also aluminum is used as metal flakes)

c) **Nanocomposites:** Nanocomposite is a multiphase solid material where one of the phases has one, two or three dimensions of less than 100 nanometers (nm), or structures having nano-scale repeat distances between the different phases that make up the material. The idea behind Nanocomposite is to use building blocks with dimensions in nanometer range to design and create new materials with unprecedented flexibility and improvement in their physical properties. Nanocomposites are found in nature, for example in the structure of the abalone shell and bone.

d) **Fiber composites :** A fiber-reinforced composite (*FRC*) is a composite building material that consists of three components:

   i. The fibers as the discontinuous or dispersed phase,
   ii. The matrix as the continuous phase and
   iii. The fine interphase region, also known as the interface.

3.4 Composite Fabrication Processes:

There are different types of Composite Fabrication process. Those are given below:

a) Vacuum Bag Molding
b) Pressure Bag Molding
c) Autoclave Molding
d) Resin transfer molding (*RTM*)
e) Open Molding
   i. Hand lay-up Process
   ii. Spray-up Process

In spite of different fabrication process, Open molding process is much preferable than other process. Because of Tooling cost for open molds is relatively low, making it possible to use this technique for short production runs. So open molding fabrication process is discussed broadly in next pages.

3.4.1 Open Molding:

Open molding is a process using a rigid, one sided which shapes only one surface of the panel.

While the opposite surface is determined by the amount of material placed upon the lower mold. Reinforcement materials can be placed manually by human or robotically. For the examples of reinforcement agent are continuous fiber forms fashioned into textile constructions and chopped fiber. The matrix is generally a resin, and can be applied with a pressure roller, a spray device or manually. This process is generally done at ambient temperature and atmospheric pressure. Two variations of open molding are Hand Lay-up and Spray-up [11].

3.4.1.1 Hand lay-up Process:

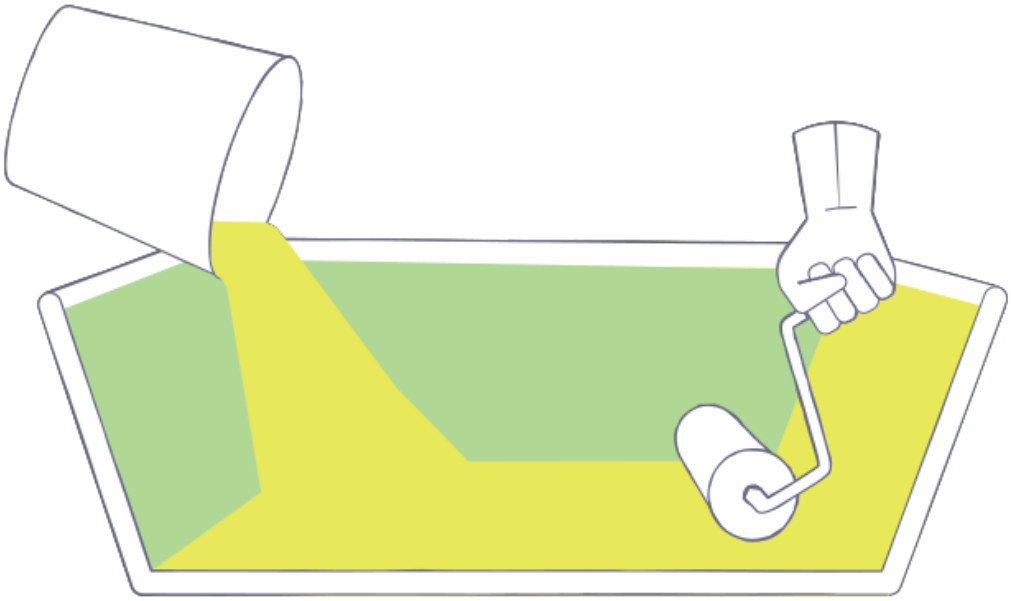


Fig. 2.5: Hand lay-up Process

Hand lay-up is the least expensive open-molding method because it requires the least amount of equipment. Fiber reinforcements are placed by hand in a mold and resin is applied with a burst or roller. This process is used to make both large and small items, including boats, storage tanks, tubs and showers.

Hand lay-up is an open molding method suitable for making a wide variety of composites products from very small to very large. Production volume per mold is low; however, it is feasible to produce substantial production quantities using multiple molds. Hand lay-up is the simplest composites molding method, offering low cost tooling, simple processing, and a wide range of part sizes. Design changes are readily made. There is a minimum investment in equipment. With skilled operators, good production rates and consistent quality are obtainable.

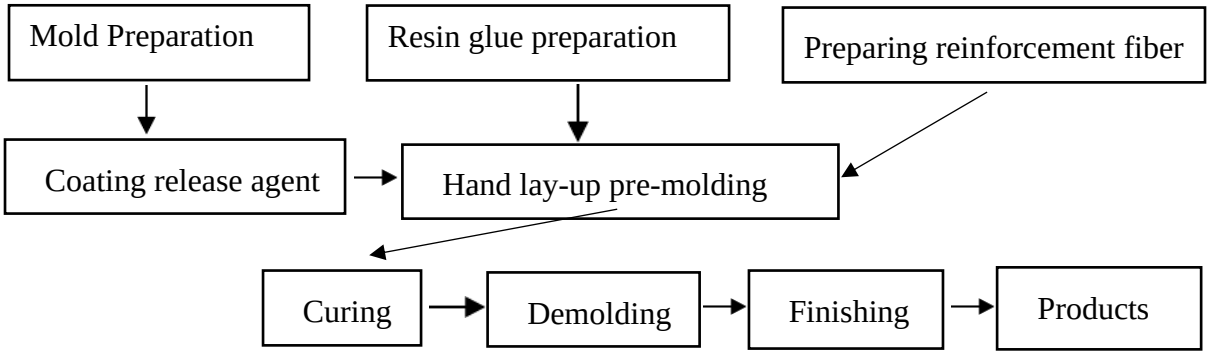


Fig. 2.5: Hand Lay-up Flowchart

3.4.1.2 Spray-up Process:

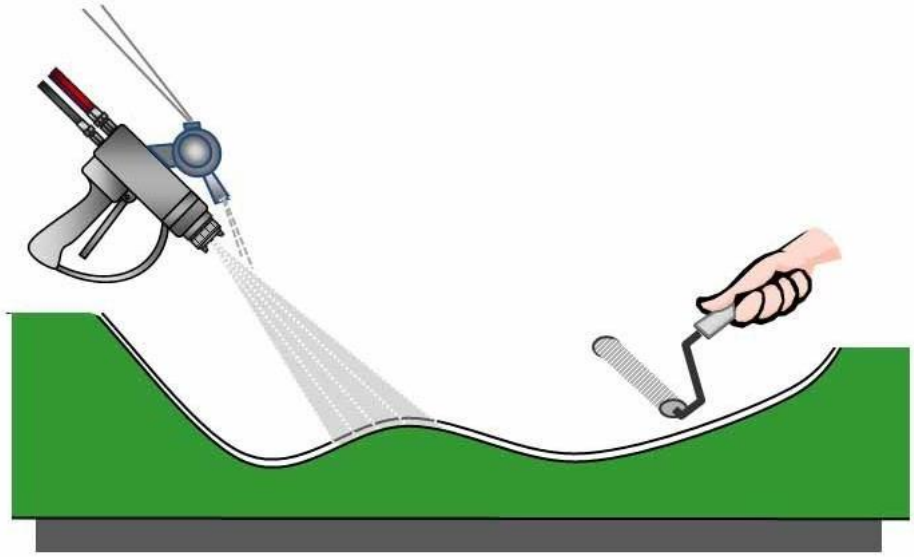


Fig. 2.6: Spray-up Process

Spray-up, or chopping, is an open mold method similar to hand lay-up in its suitability for making boats, tanks, transportation components, and tub/shower units in a large variety of shapes and sizes. A chopped laminate has good conformability and is sometimes faster to produce than a part made with hand lay-up when molding complex shapes.

In the spray-up process, the operator controls thickness and consistency, therefore the process is more operator dependent than hand lay-up. Although production volume per mold is low, it is feasible to produce substantial production quantities using multiple molds. This process uses

simple, low cost tooling and simple processing. Portable equipment permits on-site fabrication with virtually no part size limitations. The process may be automated [12].

4.0 Fabrication of Cloth

In this present work 3 types of cloth was made with 3 different material. Which are listed below:

Twill type jute mat: Twill type jute mat are made of jute yarn. There can be many types of jute yarn depending on the size of jute yarn.

Unidirectional jute mat by yarn: Jute yarn is made by raw jute fiber. Also jute yarn has different size. Unidirectional jute mat by yarn which is used in this present work is prepared by manual hand loom.

Jute mat by raw fiver: Jute is a long, soft, shiny vegetable fiber that can be spun into coarse, strong threads. It is produced primarily from plants. Jute raw fiber was bought from the market then the mat were fabricated by the manual loom process.



(a)



(b)



(c)

Fig. 4.1: (a) Jute mat (b) Raw jute fiber (c) Jute yarn

3 types of woven jute cloth were made which is given below:

i. Cloth made by jute mat.
ii. Cloth made by jute yarn.
iii. Cloth made by raw jute fiber.

Above these 3 types of cloth jute mat was bought from the market. Since jute mat and cloth is same thus jute mat was used as jute mat cloth. And jute yarn and raw jute fiber cloth was made in a handloom. A loom is a device used to weave cloth and tapestry. Figure of the handloom was used in this present work is given below.

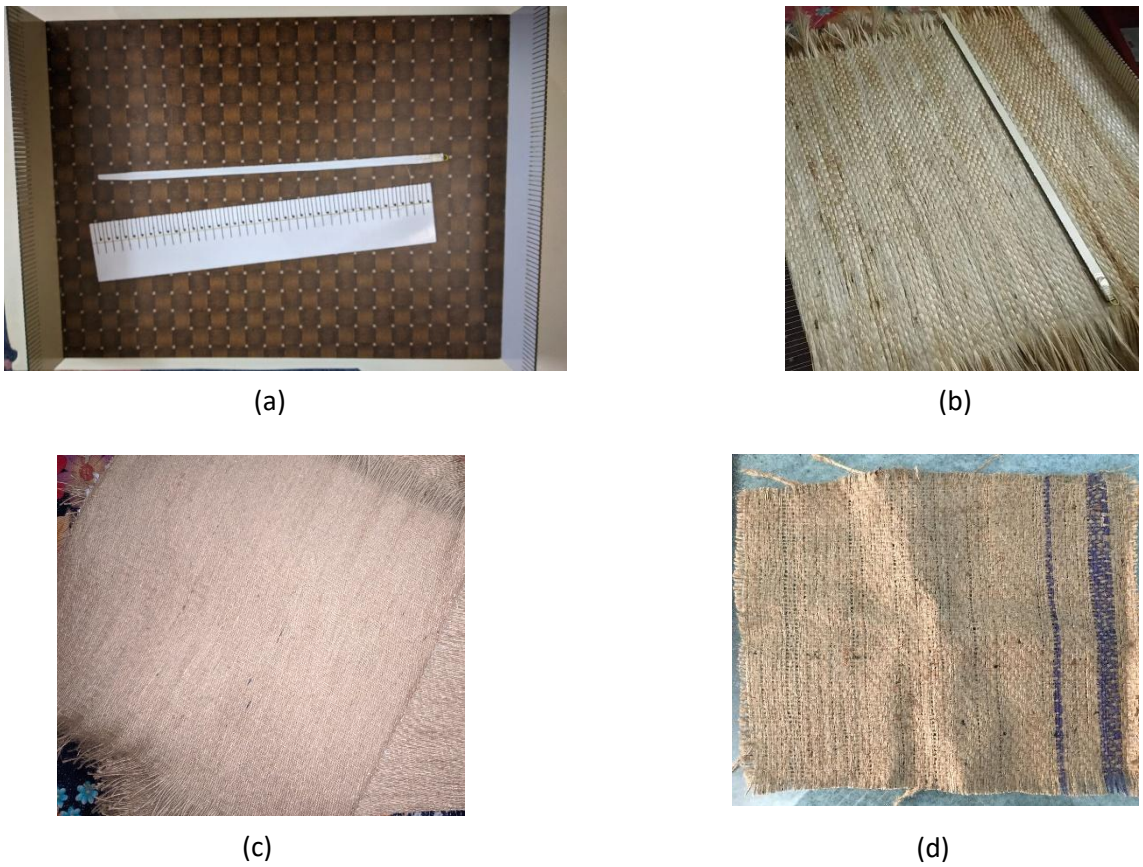


Fig. 4.2: (a) Hand loom, (b) Unidirectional cloth made by jute raw fiber, (c) Unidirectional cloth made by jute yarn (d) Twill type jute mat cloth.

4.0 Fabrication of Composites

As hand lay-up method is cheapest and easiest process of fiber reinforced composite fabrication, it is used for fabricating jute composite roofing sheet. Detailed fabrication process of jute fiber composite is given below:

4.1 Apparatus Required:

The apparatus given below are required in hand lay-up composite fabrication method:

i. Roofing steel sheet (Mould)
ii. Jute fiber

iii. Polyethylene Cover
iv. Roller
v. Hand Gloves
vi. Mug
vii. Weight meter
viii. Epoxy resin
ix. Hardener

4.2 Mold making

A roofing sheet made of steel was used as mold which was bought from the market. Casting of the composite material will be done in the mold by hand lay-up process.

4.3 Preparation Resin and Hardener:



Fig. 4.3.1: Resin and Hardener

Specific amount of Epoxy resin and hardener have to be mixed with the ratio of 11:1 by weight.

4.4 Procedure

Step 1. The epoxy resin-hardener mixer was kept ready and the cloth was prepared. The surface of the mold plate was cleaned later.

Step 2. A polythene cover was put on the roofing steel sheet then also the cloth was put on the polythene cover. Here the polythene cover acts as a releasing agent.

Step 3. The resin was poured onto the fiber and it was accomplish by a brush for uniform distribution of the resin over the fiber. And wait for some time so as to be impregnated.

Step 4. Then the composite board was cut into the pieces as per the required dimensions.



(a)



(b)



(c)

Fig 4.4.1: (a) Roofing sheet by unidirectional mat by fiber, (b) Roofing sheet by unidirectional mat by yarn, (c) Roofing sheet by twill type mat by yarn.

5.0 Mechanical property testing: Flexural Test ASTM D790, ISO 178



Fig. 5.1: Three point bending flexural test

The flexural test measures the force required to bend a beam under three point loading conditions. The data is often used to select materials for parts that will support loads without flexing. Flexural modulus is used as an indication of a material's stiffness when flexed. Since the physical properties of many materials (especially thermoplastics) can vary depending on ambient temperature, it is sometimes appropriate to test materials at temperatures that simulate the intended end use environment.

A variety of specimen shapes can be used for this test, but the most commonly used specimen size for ASTM is 3.2mm *x* 12.7mm *x* 125mm (0.125" *x* 0.5" *x* 5.0") and for ISO is 10mm *x* 4mm *x* 80mm. From the three point bending test stress and strain can be found by given formula:

$$\sigma = \frac{3FL}{2bd^2} \quad \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \quad \text{(i)}$$

$$\epsilon = \frac{6Dd}{L^2} \quad \ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots \quad \text{(ii)}$$

Where,

$\sigma =$ Stress in outer fibers at midpoint, (N/m^2)

$\epsilon =$ Strain in the outer surface, (m/m)

$F =$ Load at the midpoint of the specimen, (N)

$L =$ Length of the specimen, (m)

$b =$ Width of the specimen, (m)

$d =$ Thickness of the specimen, (m)

$D =$ Maximum deflection of the center of the specimen, (m)

Most commonly the specimen lies on a support span and the load is applied to the center by the loading nose producing three point bending at a specified rate. The parameters for this test are the support span, the speed of the loading, and the maximum deflection for the test [12].

6.0 Results and Discussion

Table 6.1: Physical parameters of fabricated roofing sheet

	Roofing sheet by unidirectional mat by yarn	Roofing sheet by unidirectional mat by fiber	Roofing sheet by twill type mat by yarn
Density, ρ (kg/m^3)	873.4947485	958.1600716	1069.584919
Specific gravity	0.873494749	0.958160072	1.069584919

From the table 6.1 density and specific gravity are seen for different roofing sheet. Also, it is seen that roofing sheet by unidirectional mat by yarn has the lowest density and specific gravity than others and roofing sheet by twill type mat by yarn has the highest density and specific gravity and roofing sheet by unidirectional mat by fiber has density and specific gravity between the other two.

Flexural test result:

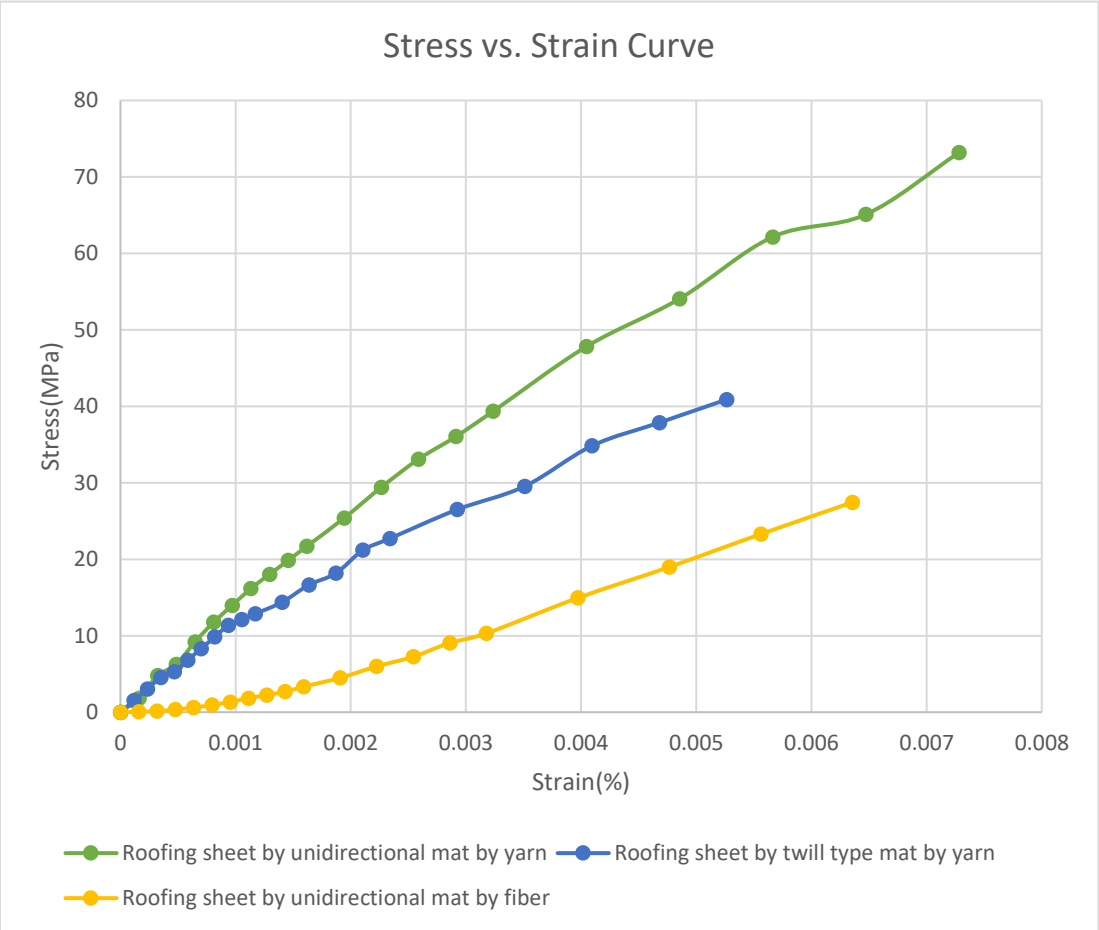


Fig. 6.1: Flexural stress-strain diagram for different composites

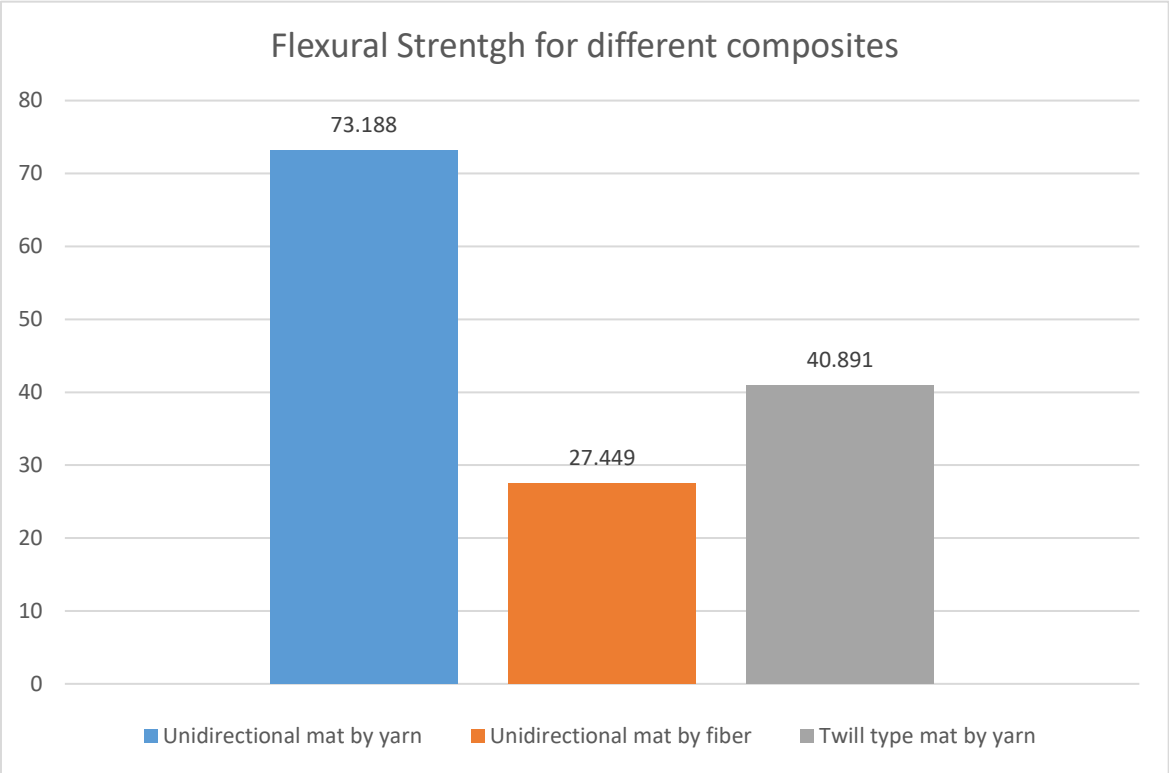


Fig. 6.2: Flexural strength of different composites

In figure 6.2, *y*-axes represent the flexural strength, σ (MPa) of different types of fabricated roofing shed. From this figure it is seen that roofing sheet made by unidirectional mat by yarn has the highest flexural strength and roofing sheet made by unidirectional mat by fiber has the lowest flexural strength.

7.0 Conclusion

The importance of materials in modern world can be realized from the fact that much of the research is being done to apply new materials to different components. From this result, it is shown that roofing sheet made by unidirectional mat by yarn gives the highest flexural strength and has the lowest density and the lowest specific gravity. So, it will give better performance in manufacturing roofing sheet. Also, as jute fiber has lower thermal conductivity than conventional roofing sheet, it will give comfortable environment during summer season.

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